```
---------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[20], line 64
     62     fig.suptitle(f'Fixed ω₀={omega0}, varying α')
     63     plt.tight_layout()
---> 64     plt.savefig(f"Fixed w_0={omega0}, varying alpha")
     65     plt.show()
     67 # ============================================================
     68 # 2) Three figures: fixed α, ω0 varies over omega_vec
     69 # ============================================================

File ~/miniconda3/envs/ecg-byte/lib/python3.10/site-packages/matplotlib/pyplot.py:1228, in savefig(*args, **kwargs)
   1225 fig = gcf()
   1226 # savefig default implementation has no return, so mypy is unhappy
   1227 # presumably this is here because subclasses can return?
-> 1228 res = fig.savefig(*args, **kwargs)  # type: ignore[func-returns-value]
   1229 fig.canvas.draw_idle()  # Need this if 'transparent=True', to reset colors.
   1230 return res

File ~/miniconda3/envs/ecg-byte/lib/python3.10/site-packages/matplotlib/figure.py:3395, in Figure.savefig(self, fname, transparent, **kwargs)
   3393     for ax in self.axes:
   3394         _recursively_make_axes_transparent(stack, ax)
```

```
-> 3395 self.canvas.print_figure(fname, **kwargs)

File ~/miniconda3/envs/ecg-byte/lib/python3.10/site-packages/matplotlib/backend_bases.py:2144, in FigureCanvasBase.print_figure(self, filename, dpi, facecolor, edgecolor, orientation, format, bbox_inches, pad_inches, bbox_extra_artists, backend, **kwargs)
   2139     _api.warn_deprecated(
   2140         "3.8", name="papertype='auto'", addendum="Pass an explicit paper type, "
   2141         "'figure', or omit the *papertype* argument entirely.")
   2143 # Remove the figure manager, if any, to avoid resizing the GUI widget.
-> 2144 with cbook._setattr_cm(self, manager=None), \
   2145 self._switch_canvas_and_return_print_method(format, backend) \
   2146         as print_method, \
   2147      cbook._setattr_cm(self.figure, dpi=dpi), \
   2148      cbook._setattr_cm(self.figure.canvas, _device_pixel_ratio=1), \
   2149      cbook._setattr_cm(self.figure.canvas, _is_saving=True), \
   2150      ExitStack() as stack:
   2152     for prop in ["facecolor", "edgecolor"]:
   2153         color = locals()[prop]

File ~/miniconda3/envs/ecg-byte/lib/python3.10/contextlib.py:135, in _GeneratorContextManager.__enter__(self)
    133 del self.args, self.kwds, self.func
    134 try:
--> 135     return next(self.gen)
    136 except StopIteration:
    137     raise RuntimeError("generator didn't yield") from None

File ~/miniconda3/envs/ecg-byte/lib/python3.10/site-packages/matplotlib/backend_bases.py:2040, in
```

```
FigureCanvasBase._switch_canvas_and_return_print_me
thod(self, fmt, backend)
  2038     canvas_class =
get_registered_canvas_class(fmt)
  2039     if canvas_class is None:
-> 2040         raise ValueError(
  2041             "Format {!r} is not supported (supported
formats: {})".format(
  2042                 fmt, ",
".join(sorted(self.get_supported_filetypes())))
  2043     canvas = canvas_class(self.figure)
  2044 canvas._is_saving = self._is_saving

ValueError: Format '283185307179586, varying alpha' is
not supported (supported formats: eps, jpeg, jpg, pdf,
pgf, png, ps, raw, rgba, svg, svgz, tif, tiff, webp)
```

The error occurs because your filename passed to `plt.savefig()` contains characters (like `=`, `,`, and spaces) that confuse Matplotlib's file format parser.
Specifically, this line:

python Copy code

```python
plt.savefig(f"Fixed w_0={omega0}, varying alpha")
```

creates a filename such as:

sql Copy code

```
Fixed w_0=3.141592653589793, varying alpha
```

Matplotlib interprets the part after the last `.` as a file format — since there's no extension, it tries to read `283185307179586, varying alpha` as the format, which causes the error.

✅ How to Fix It

Add a valid file extension (like `.png`, `.pdf`, or `.svg`) and sanitize the filename:

python Copy code

```
plt.savefig(f"Fixed_w0_{omega0:.2f}_varying_alpha.png")
```

This ensures:

- You have a valid image format (`.png`).
- Special characters and spaces are removed.
- The filename is safe and readable.

💡 Alternative (Safer Formatting)

You can also use `pathlib` to safely construct filenames:

python Copy code

```
from pathlib import Path

filename = Path(f"Fixed_w0_{omega0:.2f}_varying_alpha.png")
plt.savefig(filename)
```

✅ In Summary

Replace:

python Copy code

```
plt.savefig(f"Fixed w_0={omega0}, varying alpha")
```

with:

python Copy code

```
plt.savefig(f"Fixed_w0_{omega0:.2f}_varying_alpha.png")
```

and the error will disappear.